

automotive industry will help in decreasing the cost through the economies of scale.

Today, the SiC market is dominated by 10cm and 15cm wafers. Both of these are in high demand, but most of the growth is happening in the 15cm domain, where the aspect ratio of the wafer is 6 to 1. While the industry is looking to improve the methods of processing for 15cm wafers, companies are paying close attention to the development of 20cm SiC wafers at the same time.

The main reason for this transition is to reduce the application cost of SiC. A larger wafer size directly reduces the cost of production. But moving to a larger wafer size is not that easy and establishing mass production infrastructure takes time. Once the mass production of 20cm SiC is established, and the market is no longer dominated by just six to seven companies, the cost of SiC production will drop.

Advantages of 20cm SiC wafers

Ability to produce more SiC devices. 20cm wafers can significantly increase the production capacity of SiC devices since a larger area can be used for manufacturing. This will ensure that more devices can be produced using a single wafer of 20cm SiC. Compared to its 15cm counterpart, 20cm wafers will almost double the output of chip production.

Better wafer quality. Approximately twelve wafers are obtained in the batch production of 15cm and 10cm wafers. This number is quite less for 20cm wafers, which is why it is much easier to follow the single-wafer production approach



Buck-boost evaluation board for 650V SiC C3M MOSFET (Image source: Wolfspeed)

instead of batch production. This enables us to have better control over each wafer, in turn increasing the quality of production.

Electrification. As 20cm wafers become a norm and production capacity increases in the coming years, we can expect major growth in the electrification of systems and products, not just in the automotive sector. The industry might slowly start remaking some fabs that currently run on silicon with SiC technology.

Disadvantages of 20cm SiC wafers

Need for independent processes and increased investment costs. The production process, as well as the

development process of 20cm wafers is fundamentally different from 15cm wafers. Hence, fabrication companies will have to establish independent process plants for the 20cm technology, and this will add to their costs. Not to mention, SiC production itself is more expensive than silicon.

Risk of damages. SiC wafers are already pretty delicate. Developing a wafer of a larger area increases the risk of damaging the wafer during the substrate cutting process. Problems like cutting stress and warpage are evident as the substrate size increases.

Achieving a good tradeoff. Using 20cm SiC will make sense only if the benefits outweigh the challenges, most importantly in terms of cost. For example, the efficiency of the resulting device built on 20cm SiC should compensate for the higher SiC cost as well as the increased costs of 20cm wafer production.

Current trends in SiC industry

Mentioned below are some trends that are being witnessed currently in the silicon carbide industry:

More and more SiC products. So far, several SiC devices have been produced and many have reached maturity as well. For example, many companies, including ROHM Semiconductor, STMicroelectronics, and Infineon, have already launched several SiC based devices like MOSFETs, Schottky diodes, power modules, and hybrid devices. Also, many companies are offering commercially available evaluation kits that feature modules based on SiC.

20cm SiC wafers produced. Due to the cost benefits that the mass production of 20cm SiC will offer,

World's Leading SiC Wafer manufacturers

Many semiconductor companies are gearing up for the transition from 15cm to 20cm wafers. Here is a list of a few of the key players:

- STMicroelectronics
- Cree (Now Wolfspeed)
- II-V Incorporated
- SiCrystal (A ROHM Semiconductor Company)
- Infineon
- ON Semiconductor
- Bosch